

02 Mathematical background

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Matrices to represent data

Tables: observations in rows, features in columns

Images: grey scale

Use of matrices in ML

- Matrix equations (03 Regression)
- Matrix scaling, shifts in space (04 Preprocessing)
- Norms and distance (01 kNN, 03 Regression, 05 Tree-Based Models, 11 Clustering)
- Inner product (07 Logistic regression)
- Matrix-vector multiplication (07 Logistic Regression, 09 Neural Networks)
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Linear Equations

Variables $x_1, \dots, x_n$

Coefficients $a_1, \dots, a_n$

Linear equation: $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$

Linear equations represent straight lines, planes, and hyperplanes in various dimensions.

Systems of Linear Equations

= collection of one or more linear equations involving the same set of variables.

$$2x_1 + 3x_2 + x_3 = 3$$

$$7x_2 - 4x_3 = 10$$

$$x_3 = 1$$

Solutions to a System of Linear Equations

A **solution** of a linear system in the variables $x_1, x_2, \dots, x_n$ is a list of numbers $s_1, s_2, \dots, s_n$ that satisfies all equations of the system when substituted for the variables $x_1, x_2, \dots, x_n$, respectively.

The set of all possible solutions is called its **solution set**.

Two linear systems are called **equivalent** if they have the same solution set.

Solutions to a System of Linear Equations

- No solutions: inconsistent
 - One solution: unique set of values satisfying all equations
 - Infinitely many solutions: multiple sets of values
-
- Consistent: a solution exists (one or infinite)
 - Inconsistent: no solutions

Representing a system with a matrix

Linear system:

$$2x_1 + 3x_2 + x_3 = 3$$

$$7x_2 - 4x_3 = 10$$

$$x_3 = 1$$

Coefficient matrix

Augmented matrix

Size of a matrix

The size of a matrix is defined by the number of its rows and columns, expressed as *rows* $\times$ *columns*.

$$\begin{bmatrix} 2 & 3 & 1 \\ 0 & 7 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 3 \\ 0 & 7 & -4 & 10 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

Elementary row operations

Fundamental transformations used to solve equation systems.

- Replacement: replacing one row by the sum of itself and a multiple of another row ($r_1 \rightarrow r_1 + 3r_2$)
- Swap: swap two rows ($r_1 \leftrightarrow r_2$)
- Scaling: multiply a row by $k, k \neq 0$ ($r_1 \rightarrow -2r_1$)

Two matrices are **row equivalent** if one can be transformed into the other through a sequence of elementary row operations.

Echelon Form (EF)

A matrix is in **echelon form** iff:

- All zero rows are at the bottom.
- Each leading entry of a row is in a column to the right of the leading entry of the row above it.
- All entries in a column below a leading entry are zeros.

Reduced Row Echelon Form (RREF)

A matrix is in **reduced row echelon form** if it is in EF and:

- The leading entry in each non-zero row is 1.
- Each leading 1 is the only non-zero entry in its column.

Each matrix is row equivalent to one and only one reduced echelon matrix.

EF and RREF

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 7 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 4 & 0 & 2 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Pivots and free variables

- Pivot position: leading 1 in RREF. Pivot columns = basic variables.
- Free variables: non-pivot columns.

$$\begin{bmatrix} 1 & 4 & 0 & 2 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Uniqueness theorem

- Unique solution: no free variables
- Infinitely many solutions: at least one variable

Consistency: EF of augmented matrix has **no** row of form

$$\begin{bmatrix} 0 & 0 & 0 & b \end{bmatrix}$$

where $b \neq 0$

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Vectors

Vectors have **magnitude** and **direction**.

$\bar{v} = (x_1, x_2)$ describes movement from the origin $(0, 0)$ to the point (x_1, x_2) .

Vectors in $\mathbb{R}^n$ are often represented as **column matrices** of size $(n \times 1)$.

Vector addition and scalar multiplication

Entry-by-entry:

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

$$c\mathbf{u} = c \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} cu_1 \\ cu_2 \\ \vdots \\ cu_n \end{bmatrix}$$

Linear Combinations

Given a set of vectors $\bar{v}_1, \bar{v}_2, \dots, \bar{v}_p \in \mathbb{R}^n$ and scalars $c_1, c_2, \dots, c_p \in \mathbb{R}$, the vector $\bar{y}$ given by:

$$\bar{y} = c_1 \bar{v}_1 + \dots + c_p \bar{v}_p$$

is called a **linear combination** of $\bar{v}_1, \bar{v}_2, \dots, \bar{v}_p$ with **weights** $c_1, c_2, \dots, c_p$.

- The weights c_i can be any real numbers, including zero.
- This operation combines *scalar multiplication* and *vector addition*.

Example: linear combinations

$$\text{Let } \bar{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ and } \bar{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

$$0\bar{v}_1 + 0\bar{v}_2 =$$

$$\bar{v}_1 + 2\bar{v}_2 =$$

$$-2\bar{v}_1 - 3\bar{v}_2 =$$

$$2\bar{v}_1 + 3\bar{v}_2 =$$

Span

If $\bar{v}_1, \dots, \bar{v}_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\bar{v}_1, \dots, \bar{v}_p$ is denoted by **Span** $\{\bar{v}_1, \dots, \bar{v}_p\}$.

$$\bar{y} = c_1 \bar{v}_1 + c_2 \bar{v}_2 + \cdots + c_p \bar{v}_p$$

with $c_1, \dots, c_p$ as any scalars.

Let $\bar{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\bar{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and $\bar{v}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

$$\text{Span}\{\bar{v}_1, \bar{v}_2, \bar{v}_3\} =$$

The Matrix Equation $A\bar{x} = \bar{b}$

If A is an $m \times n$ matrix, with columns $\bar{a}_1, \dots, \bar{a}_n$, and if $\bar{x}$ is in $\mathbb{R}^n$, then the product of A and $\bar{x}$, denoted by $A\bar{x}$, is the **linear combination** of the columns of A using the corresponding entries in $\bar{x}$ as weights.

$$A\bar{x} = \bar{b}$$

$$A\bar{x} = \begin{bmatrix} \bar{a}_1 & \bar{a}_2 & \dots & \bar{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\bar{a}_1 + x_2\bar{a}_2 + \dots + x_n\bar{a}_n$$

Row-Vector Rule for Computing $A\bar{x}$

If the product $A\bar{x}$ is defined, then the i -th entry in $A\bar{x}$ is the sum of the products of corresponding entries from row i of A and from the vector $\bar{x}$.

$$\begin{bmatrix} 1 & -1 & 4 \\ 5 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} =$$

Linear independence

A set of vectors is **linearly independent** if the vector equation

$$x_1 \bar{v}_1 + \dots + x_p \bar{v}_p = \bar{0}$$

has only the **trivial solution** $(x_1, \dots, x_p) = (0, \dots, 0)$.

Otherwise, the vectors are said to be linearly dependent.

Linear independence

- If a set of vectors contains the **zero vector**, then the set is linearly dependent.
- If a set of vectors contains a **scalar multiple** of another vector, then the set is linearly dependent.
- If a set of vectors contains **more vectors** than there are **entries** in each vector, then the set is linearly dependent.

Bases in Vector Spaces

A **basis** of a vector space V is a set of vectors such that:

- The vectors are linearly independent,
- The vectors span $\mathbb{R}^n$.

The Norm

Norm = length of a vector.

Common norms:

- Euclidean norm (L_2 norm): $\|\vec{v}\|_2 = \sqrt{\sum_{i=1}^n v_i^2}$
- Manhattan norm (L_1 norm): $\|\vec{v}\|_1 = \sum_{i=1}^n |v_i|$

Normalization = process of making the length of a vector 1.

Inner product

Inner product = dot product. Measures similarity between two vectors.

$$\langle \vec{u}, \vec{v} \rangle = \sum_{i=1}^n u_i v_i$$

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Matrix addition and subtraction

Matrices must be the same size. We go entry-by-entry.

$$A + B = B + A$$

$$(A + B) + C = A + (B + C)$$

$$A + 0 = A$$

$$A + (-A) = 0$$

$$A - B = A + (-B)$$

$$A + B = A + C \Leftrightarrow B = C$$

Matrix scaling

Multiplying each entry by a scalar.

$$S(A + B) = SA + SB$$

$$(A + B)S = SA + SB, S \in \mathbb{R}$$

$$(R + S)A = RA + SA, S \in \mathbb{R}$$

$$R(SA) = (RS)A$$

Matrix multiplication

$$A : m \times n$$

$$\begin{bmatrix} x & x & x & x & x \\ x & x & x & x & x \\ x & x & x & x & x \end{bmatrix}$$

$$B : k \times l$$

$$\begin{bmatrix} x & x \\ x & x \\ x & x \\ x & x \\ x & x \end{bmatrix}$$

$A \times B$ is only defined if $n = k$: **column** of the first matrix must match the **row** of the second matrix.

$B \times A$ is **not** defined.

Size of AB is $m \times l$.

Examples

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}, C = \begin{bmatrix} 2 & 4 \\ 0 & -1 \end{bmatrix}$$

$$2B =$$

$$B + C$$

$$A \times B$$

Matrix transposition

If A is a $m \times n$ matrix, then the transpose of A , A^T , is the $n \times m$ matrix whose columns are formed from the rows of A .

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(rA)^T = rA^T, r \in \mathbb{R}$$

$$(AB)^T = B^T A^T$$

Examples: transposition

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \\ 7 & 8 \end{bmatrix}$$

$$A^T =$$

Matrix inverse

We say that a square matrix is invertible (= non-singular) iff

$$A^{-1} \cdot A = I \text{ and } A \cdot A^{-1} = I$$

where A^{-1} is the inverse of A .

Matrix inverse

If A is invertible, then $A\bar{x} = \bar{b}$ has a **unique** solution.

$$A\bar{x} = \bar{b}$$

$$A^{-1} \cdot A\bar{x} = A^{-1} \cdot \bar{b}$$

$$\bar{x} = A^{-1} \cdot \bar{b}$$

Invertible Matrix Theorem (IMT)

- (a) A is invertible.
- (b) A is row equivalent to I_n .
- (c) A has n pivot positions (i.e. one for each row and column).
- (d) The equation $A\bar{x} = \bar{0}$ has only the trivial solution.
- (e) The columns of A are linearly independent.
- (f) The equation $A\bar{x} = \bar{b}$ has a unique solution for each $\bar{b} \in \mathbb{R}^n$.
- (g) The columns of A span $\mathbb{R}^n$.
- (h) $\det(A) \neq 0$.
- (i) There is an $n \times n$ matrix C such that $CA = I$.
- (j) There is an $n \times n$ matrix D such that $AD = I$.
- (k) A^T is invertible.

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Eigenvalues, eigenvectors

An **eigenvector** is a non-zero vector $\bar{x}$ such that $A\bar{x} = \lambda\bar{x}$, $\lambda \in \mathbb{R}$.
A scalar λ is an **eigenvalue** of A if there is a non-trivial solution $\bar{x}$ of $A\bar{x} = \lambda\bar{x}$.

In such cases we say that $\bar{x}$ is an eigenvector corresponding to λ .
 $T : \bar{x} \rightarrow A\bar{x}$.

λ is an eigenvalue of A iff $A - \lambda I = 0$ has a non-trivial solution.

Eigenvalue decomposition

If A has n linearly independent eigenvectors, it can be diagonalized as:

$$A = PDP^{-1}$$

P : matrix of eigenvectors,

D : diagonal matrix with eigenvalues as entries,

P^{-1} : inverse of P .

Singular Value Decomposition (SVD)

Matrix factorization:

$$A = U\Sigma V^T$$

U : orthogonal matrix, columns are left-singular vectors of A ,

V : orthogonal matrix, columns are right-singular vectors of A ,

Σ : diagonal matrix, singular values as entries

Decomposition of transformation into rotation, scaling and rotation.

Use of SVD

- Image compression,
- Dimensionality reduction (PCA),
- Solving linear systems,
- Low-rank approximation,
- ...

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Python examples

Github link: https://github.com/304125/MAL1_S26/tree/master/02_Mathematical_Background